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Roll no- 08

Batch- T1

```
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
# 1. Load Titanic dataset
file_path = r"C:\Users\PVGIT\Downloads\archive (8).zip"
df = pd.read_csv(file_path)
print("First 5 rows:")
print(df.head()
# 2. Data Cleaning
print("\nDataset Information:")
df.info()
# 3. Handle Missing Values
df["Age"] = df["Age"].fillna(df["Age"].mean())
df["Fare"] = df["Fare"].fillna(df["Fare"].mean())
# Fill missing Embarked values with mode
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
print("\nMissing values:")
print(df.isnull().sum())
# 4. Encoding
# Convert Sex into numerical values
df["Sex"] = df["Sex"].map({
    "male": 0,
    "female": 1
})
# Convert Embarked into numerical values
df["Embarked"] = df["Embarked"].map({
```

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    "S": 0,
    "C": 1,
    "Q": 2
})
print("\nDataset after Encoding:")
print(df.head())
# 5. Statistical Analysis
print("\nStatistical Analysis:")
print(df.describe())
# 6. Select Input and Output
X = df[[
    "Pclass",
    "Sex",
    "Age",
    "SibSp",
    "Parch",
    "Fare",
    "Embarked"
]]
y = df["Survived"]
# 7. Split Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)
print("\nTraining data:", X_train.shape)
print("Testing data:", X_test.shape)

# 8. Feature Scaling
```

```
scaler = StandardScaler()
X_train = scaler.fit_transform(X_train)
X_test = scaler.transform(X_test)
print("\nScaled Training Data:")
print(X_train[:5])
# 9. Histogram
df.hist(figsize=(10, 8))
plt.suptitle("Histograms")
plt.tight_layout()
plt.show()
# 10. Boxplot
plt.figure(figsize=(8, 5))
df[[
    "Age",
    "Fare",
    "SibSp",
    "Parch"
]].boxplot()
plt.title("Boxplot")
plt.ylabel("Values")
plt.show()
```

OUTPUT-

```
First 5 rows:
   PassengerId  Survived  Pclass  \
0          892         0       3
1          893         1       3
2          894         0       2
3          895         0       3
4          896         1       3
```

```
                                          Name     Sex   Age  SibSp  Parch  \
0                                Kelly, Mr. James    male  34.5      0      0
1                    Wilkes, Mrs. James (Ellen Needs)  female  47.0      1      0
2                           Myles, Mr. Thomas Francis    male  62.0      0      0
3                                Wirz, Mr. Albert    male  27.0      0      0
4  Hirvonen, Mrs. Alexander (Helga E Lindqvist)  female  22.0      1      1

   Ticket     Fare Cabin Embarked
0  330911   7.8292   NaN        Q
1  363272   7.0000   NaN        S
2  240276   9.6875   NaN        Q
3  315154   8.6625   NaN        S
4 3101298  12.2875   NaN        S

Dataset Information:
<class 'pandas.DataFrame'>
RangeIndex: 418 entries, 0 to 417
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  418 non-null    int64
 1   Survived     418 non-null    int64
 2   Pclass       418 non-null    int64
 3   Name         418 non-null    str
 4   Sex          418 non-null    str
 5   Age          332 non-null    float64
 6   SibSp        418 non-null    int64
 7   Parch        418 non-null    int64
 8   Ticket       418 non-null    str
 9   Fare         417 non-null    float64
 10  Cabin        91 non-null     str
```

```
 11  Embarked     418 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 56.1 KB

Missing values:
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          327
Embarked         0
dtype: int64

Dataset after Encoding:
   PassengerId  Survived  Pclass  \
0          892         0       3
1          893         1       3
2          894         0       2
3          895         0       3
4          896         1       3

                                          Name  Sex   Age  SibSp  Parch  \
0                                Kelly, Mr. James    0  34.5      0      0
1                Wilkes, Mrs. James (Ellen Needs)    1  47.0      1      0
2                       Myles, Mr. Thomas Francis    0  62.0      0      0
```

3 Wirz, Mr. Albert 0 27.0 0 0

4 Hirvonen, Mrs. Alexander (Helga E Lindqvist) 1 22.0 1 1

	Ticket	Fare	Cabin	Embarked
0	330911	7.8292	NaN	2
1	363272	7.0000	NaN	0
2	240276	9.6875	NaN	2
3	315154	8.6625	NaN	0
4	3101298	12.2875	NaN	0

Statistical Analysis:

	PassengerId	Survived	Pclass	Sex	Age \
count	418.000000	418.000000	418.000000	418.000000	418.000000
mean	1100.500000	0.363636	2.265550	0.363636	30.272590
std	120.810458	0.481622	0.841838	0.481622	12.634534
min	892.000000	0.000000	1.000000	0.000000	0.170000
25%	996.250000	0.000000	1.000000	0.000000	23.000000
50%	1100.500000	0.000000	3.000000	0.000000	30.272590
75%	1204.750000	1.000000	3.000000	1.000000	35.750000
max	1309.000000	1.000000	3.000000	1.000000	76.000000

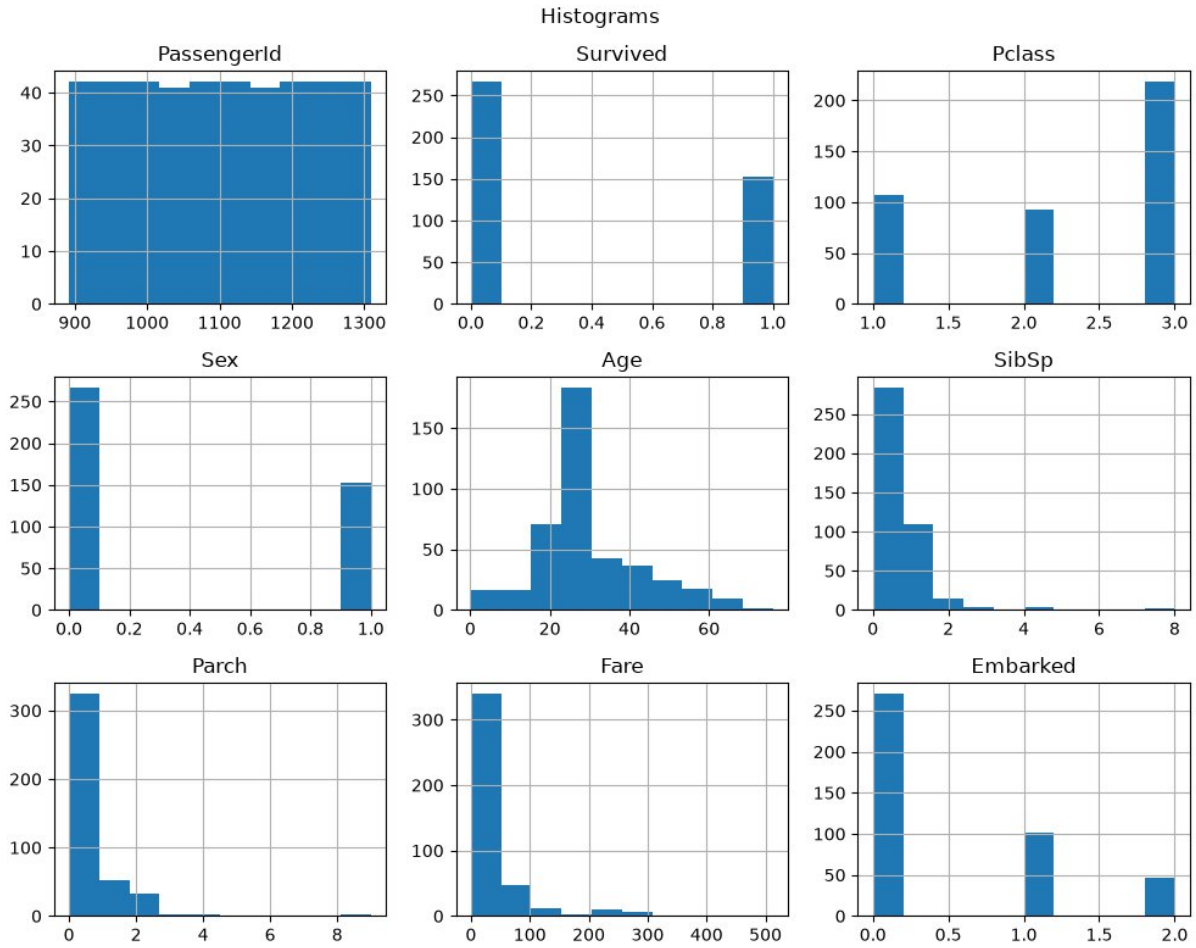
	SibSp	Parch	Fare	Embarked
count	418.000000	418.000000	418.000000	418.000000
mean	0.447368	0.392344	35.627188	0.464115
std	0.896760	0.981429	55.840500	0.685516
min	0.000000	0.000000	0.000000	0.000000
25%	0.000000	0.000000	7.895800	0.000000
50%	0.000000	0.000000	14.454200	0.000000
75%	1.000000	0.000000	31.500000	1.000000
max	8.000000	9.000000	512.329200	2.000000

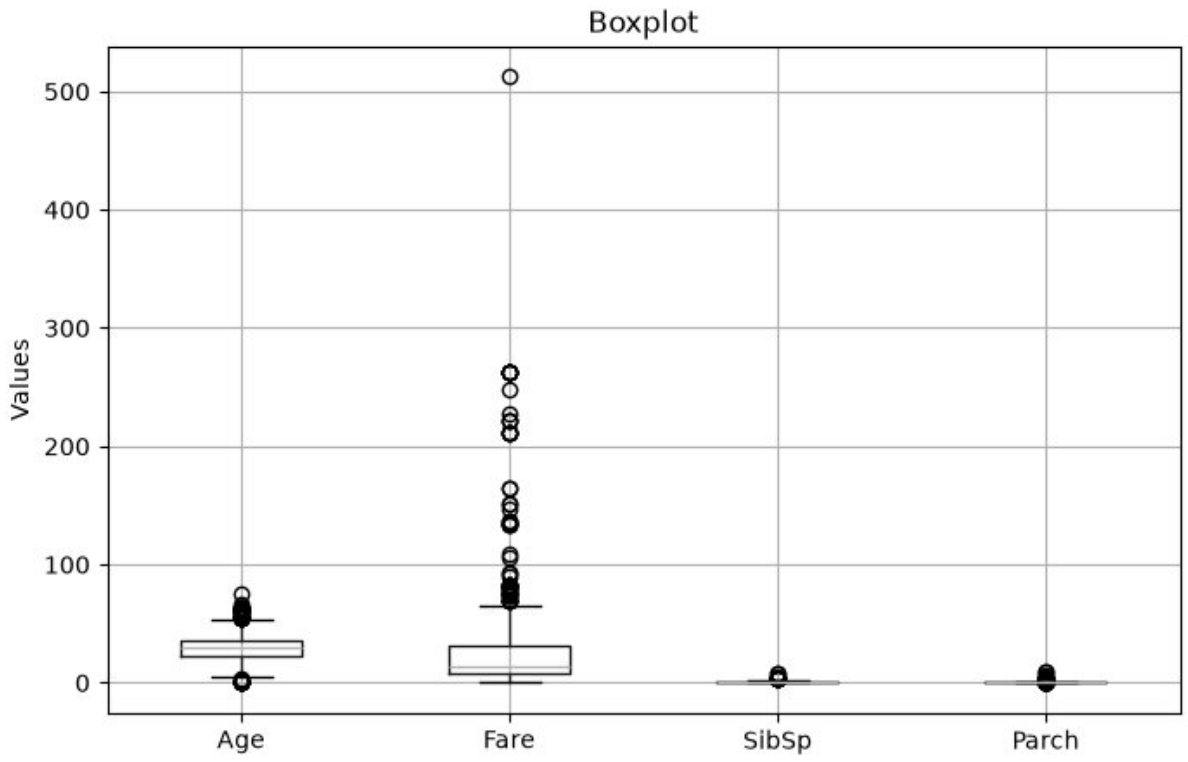
Training data: (334, 7)

Testing data: (84, 7)

Scaled Training Data:

[[-0.31938196 -0.73911859 0.14282735 -0.49831227 -0.43196296 -0.41301013
  -0.68506839]
 [-0.31938196 -0.73911859 -0.47466103 1.62189534 -0.43196296 -0.09338694
  -0.68506839]
 [-0.31938196 -0.73911859 0.00949543 -0.49831227 -0.43196296 -0.45260367
   2.21129669]
 [-1.5046439 -0.73911859 -0.47466103 0.56179154 -0.43196296 0.78370577
  -0.68506839]
 [-0.31938196 -0.73911859 -0.86059127 -0.49831227 -0.43196296 -0.45620246
  -0.68506839]]




Boxplot
Values
500
400
300
200
100
0
Age
Fare
SibSp
Parch